

Research Project Timeline

Stage	Activity	Estimated duration	Start date	End date	Deliverables	Comments/Notes
Research Design and Planning	- Finalize research problem/questions - Develop research design - Prepare research proposal	4 Weeks	29 Sept 2025	26 Oct 2025	• Finalized research questions and objectives • Research proposal slides • Approve research plan/timeline	Spend ~1 week each on: FinBERT/FinGPT, sentiment in finance, hybrid prediction models, Malaysian studies. End with a clear conceptual framework.
Literature Review	- Search, capture and synthesize relevant literature - Prepare draft literature review	3 Weeks	27 Oct 2025	15 Nov 2025	• Annotated bibliography of 10-15 key papers • Draft literature review chapter/section • Summary table of key studies (FinBERT, FinGPT, Malaysian stock prediction)	
Data Collection and Preparation	- Scrape financial/political news (The Star, NST, Malaysiakini, etc.) - Collect KLSE stock data - Annotate sample news heading (1-2k items) - Clean and preprocess datasets	3 Weeks	17 Nov 2025	7 Dec 2025	• Cleaned and merged datasets: - news_sentiment.csv (labeled Malaysian financial/political news) - stock_data.csv (KLSE stock data with indicators) • Documentation of data sources and preprocessing steps • Small labeled subset for validation (1-2k samples)	Annotation is time-consuming, plan for 1-2 hrs/day labelling. Ensure scripts for scraping and preprocessing are reusable.
Experiment 1 – Sentiment Model Benchmarking	- Run FinBERT & FinGPT (zero-shot and fine-tuned) - Evaluate with metrics (F1, precision, recall) - Document errors & qualitative differences	3 Weeks	8 Dec 2025	21 Dec 2025	• Evaluation report comparing FinBERT vs FinGPT (F1, precision, recall) • Confusion matrices and bar charts of performance • Summary table of qualitative examples • Jupyter notebook/code for reproducibility	Spend ~ 1 week per model setup and 1 week analyzing & reporting results.
Experiment 2 – Sentiment vs Stock Movement	- Merge daily sentiment with stock returns - Perform correlation, regression, and Granger causality - Visualize and interpret results	2 Weeks	22 Dec 2025	4 Jan 2026	• Correlation, regression, and Granger causality results • Visuals: lag correlation heatmap, sentiment vs price plots • Analytical report describing relationship strength and interpretation	Focus on clean data alignment (date matching). Automate feature aggregation (daily averages).

Experiment 3 – Stock Price Prediction Models	- Build LSTM/GRU baseline models - Add sentiment-enhanced features - Implement Transformer with FinGPT embeddings - Compare results (RMSE, MAE, accuracy)	2 Weeks	5 Jan 2026	18 Jan 2026	• Trained LSTM, GRU and Transformer models • Model performance comparison table (RMSE, MAE, directional accuracy) • Visualization: predicted vs actual price trends • Technical appendix/code documentation	Most technical phase/ Plan 2 weeks for model training & tuning, 2 weeks for evaluation and visualization.
Synthesis, Analysis and Writing	- Integrate all results - Discuss contributions, limitations, and future work - Final draft of report - Review draft with supervisor - Final editing	2 Weeks	19 Jan 2026	31 Jan 2026	• Integrated discussion and interpretation of findings • Final draft of research report • Abstract, introduction, and conclusion chapters polished • Supervisor feedback log and final edited version	Aim for a structured write-up. Iterate drafts with supervisor feedback